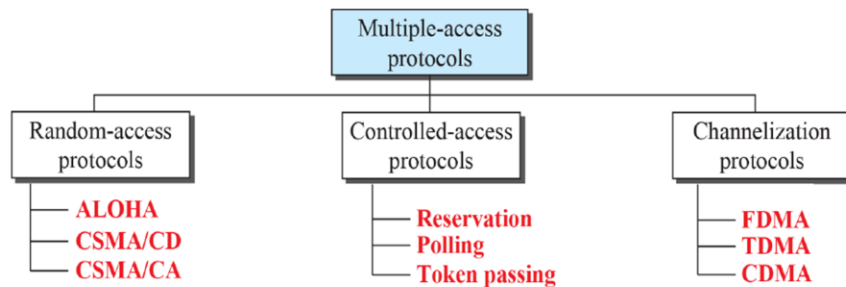


Multiple-access protocols

Taxonomy of multiple-access protocols



Multiple Access Control –

If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously. Hence multiple access protocols are required to decrease collision and avoid crosstalk.

RANDOM ACCESS

- or contention
- at each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
- this decision depends on the state of the medium (idle or busy).

Random Access Protocol:

In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state (idle or busy). It has two features:

There is no fixed time for sending data

There is no fixed sequence of stations sending data

Primary Purpose of Random Access

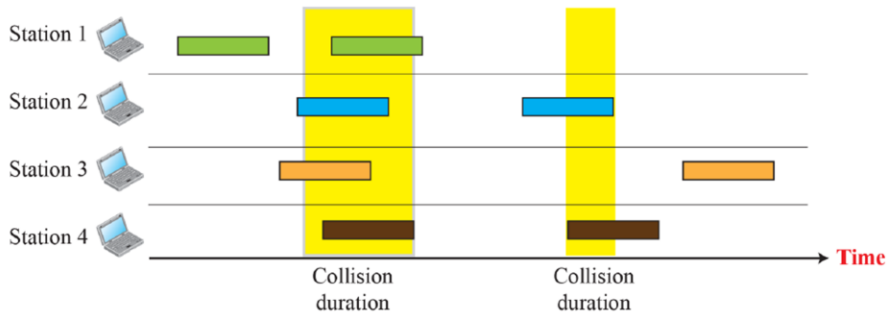
- The primary purpose of Random Access protocols is to facilitate shared access to a communication medium among multiple network devices, allowing them to transmit data without prior coordination or scheduling.

ALOHA

- **-the earliest random access method was developed in early 1970.**
- **Pure ALOHA**
- **Slotted ALOHA**

(a) ALOHA – In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

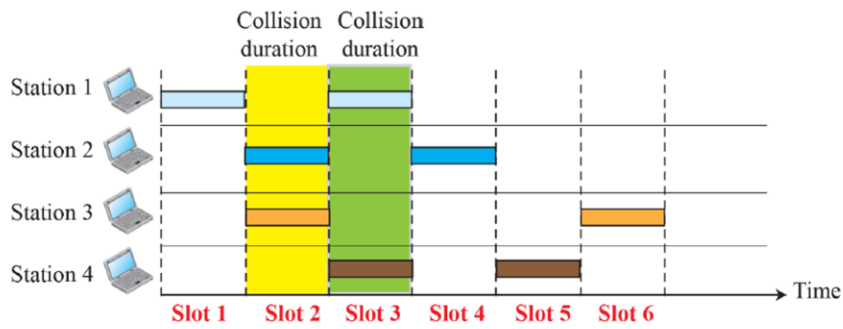
Frames in a pure ALOHA network



- Pure Aloha:

When a station sends data it waits for an acknowledgement. If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (T_b) and re-sends the data. Since different stations wait for different amount of time, the probability of further collision decreases.

Frames in a slotted ALOHA network



Slotted Aloha:

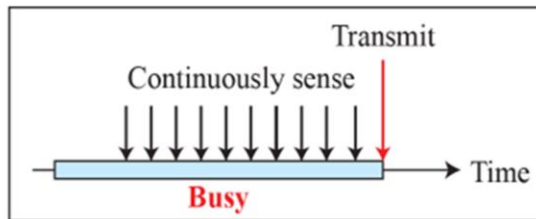
It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots. If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

CSMA

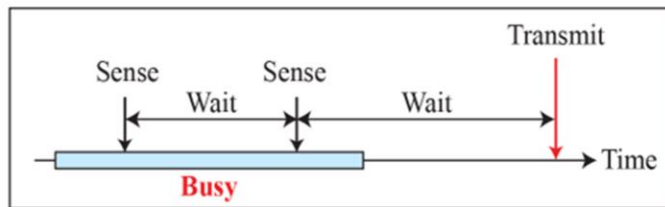
- - The chance of collision can be reduced if a station senses the medium before trying to use it.
- - Carrier sense multiple access (CSMA) requires that each station first listen to the medium (or check the state of the medium) before sending.
- - CSMA is based on the principle
“sense before transmit” or
“listen before talk.”

CSMA – Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data. If it is idle then it sends data, otherwise it waits till the channel becomes idle. However there is still chance of collision in CSMA due to propagation delay. For example, if station A wants to send data, it will first sense the medium. If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.

Persistent Methods



a. 1-persistent



b. Nonpersistent

CSMA access modes-

1-persistent: The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally (with 1 probability) as soon as the channel gets idle.

Non-Persistent: The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.

CSMA/CD

- a station monitors the medium after it sends a frame to see if the transmission was successful.
- If so, the station is finished. If, however, there is a collision, the frame is sent again.

Key Points

- **Carrier Sense:** Before transmitting data, a device using CSMA/CD listens for carrier signals on the network to determine if the channel is busy or idle. If the channel is sensed as busy, the device waits until it becomes idle before transmitting.
- **Multiple Access:** Multiple devices share the same network channel. Any device that needs to transmit data must first check if the channel is available for transmission.
- **Collision Detection:** If two devices begin transmitting data at nearly the same time, a collision may occur. Devices continue to monitor the channel while they transmit to detect collisions.
- **Backoff and Retransmission:** When a collision is detected, the devices involved stop transmitting and wait for a random period of time before reattempting transmission. This is known as the "backoff" period. The randomization helps reduce the likelihood of another collision during retransmission attempts.

CSMA/CA

Key Points

- **Carrier Sense:** Similar to CSMA/CD, CSMA/CA devices listen for carrier signals on the wireless channel to determine if it's busy or idle before attempting transmission.
- **Multiple Access:** Multiple devices share the same channel. Devices use CSMA/CA to decide when to transmit data to minimize the risk of collisions.
- **Collision Avoidance:** Instead of detecting collisions after they occur, CSMA/CA aims to avoid collisions altogether. Devices implement a mechanism to defer transmission if they sense the channel as busy.
- **Random Backoff:** When a device wants to transmit data, it waits for a random backoff period before attempting transmission. The randomization helps avoid synchronized collisions that might occur if devices started transmitting simultaneously after sensing the channel as idle.
- **Acknowledgment (ACK):** After a device transmits a frame, the receiving device sends an acknowledgment (ACK) frame to confirm successful reception. If the sender doesn't receive an ACK within a specified time, it assumes a collision occurred and retransmits.
- **Request-to-Send (RTS) and Clear-to-Send (CTS):** In CSMA/CA, devices can use RTS and CTS frames to signal their intention to transmit data.